



THESIS ECONOMETRICS AND DATA SCIENCE

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Evaluating the RVU levy exemption using AI generated survey responses

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Abstract

This thesis evaluates the RVU levy exemption, a regulation designed to help individuals with demanding jobs retire early. As data on demanding work is not readily available for users of the exemption, we estimate the fraction of exemption users who had a demanding job by predicting responses to several questions related to demanding work. To this end, we select six questions related to different aspect associated with demanding work in the LISS Work & Schooling survey, convert these to binary indicators, and train a machine learning model that predicts responses based on features in the CBS microdata. To accurately estimate the relation between demanding work and exemption usage, we estimate misclassification rates for several subpopulations. Under the unlikely assumption of nondifferentiality of prediction errors, individuals who had demanding jobs make up a significant part of exemption users and they appear overrepresented compared to both the employed population and the exemption-eligible population. A robustness analysis shows that these results depend on the validity of the nondifferentiality assumption, so care must be taken with these claims.

Acknowledgements

I would like to thank LISS, CBS, UWV and SEO

LISS statements

In this paper, we make use of data from the LISS panel (Longitudinal Internet studies for the Social Sciences), which is administered and managed by the non-profit research institute Centerdata (Tilburg University, the Netherlands).

The LISS panel data were collected by the non-profit research institute Centerdata (Tilburg University, the Netherlands). Funding for the panel's ongoing operations comes from the Domain Plan SSH and ODISSEI since 2019. The initial set-up of the LISS panel in 2007 was funded through the MESS project by the Netherlands Organization for Scientific Research (NWO).

The LISS panel is a representative sample of Dutch individuals who participate in monthly internet surveys. The panel is based on a true probability sample of households drawn from the population register by Statistics Netherlands. Self-registration is not possible, and households that would otherwise be unable to participate are provided with a computer and internet connection. The longitudinal LISS Core Study, consisting of separate topical surveys, has been conducted in the panel every year since 2007, covering a large variety of domains including health, work, education, income, housing, leisure and time use, political views, values and personality.

CBS

CBS moet ik ook bedanken

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1 Introduction

1.1 Motivation & research question

To increase labour market participation of older citizens, in recent years the statutory retirement age (SRA) in the Netherlands, also referred to as the AOW age, has increased stepwise from 65 to 67 years and is planned to increase further in the future (bron nieuw artikel). In addition, early retirement policies have become stricter, giving fewer people the opportunity to retire before the SRA (Visser et al., 2016).

Where these policies help reduce public costs and increase public income, concerns have been raised about people with demanding jobs. Many already struggled to reach retirement at the old SRA in good health (Vermeer et al., 2016) and some entered social support programs (Chirikos and Nestel, 1991). With recent (and future) increases, this problem is expected to get worse (source).

The Netherlands has had several early retirement policies over the years that supported people nearing the SRA in retiring early. Recently, in 2021, a new (temporary) early retirement policy was introduced in the form of the RVU levy exemption. This regulation offers the possibility of fiscal benefits when retiring up to 3 years early and is mainly aimed at supporting individuals with heavy work (bron, webpagina). However, because it is hard to classify which jobs are "demanding", employers and employees can agree to use the exemption without proving that the employee in question does demanding work.

As everyone can make use of the RVU levy exemption, it has to be assessed whether the target audience is reached or whether people with non-demanding jobs are using it to retire early. If the latter is the case, this could cost the public substantial sums, and hence it is important to assess whether the exemption has the intended effects.

To my knowledge, there is no existing research regarding the effects of the RVU levy exemption, and it has not been investigated whether the regulation works as intended. Therefore, this thesis aims to answer the question:

How effective is the RVU levy exemption to support people with demanding jobs with early retirement?

The difficulty in answering this question is twofold. Firstly, there is no one clear definition of what jobs are classified as "demanding", and there are different aspects that can make work demanding, physically or mentally (reference). Secondly, even when agreeing on which aspects make a job demanding, there is no readily available data on these aspects for individuals who used the RVU levy exemption.

1.2 Procedure & problems

To account for these difficulties, we use a machine learning model to predict several aspects of demanding work for individuals who used the exemption. In particular, as a proxy, we use responses to questions such as "Did/do you work irregular hours?" and "Did/do you need to lift heavy objects?" in the LISS panel Work & Schooling survey. As features, we use several individual-level datasets in the CBS microdata, including education level, gender, and job-related information such as salary, extra hours worked, and the sector in which they worked.

Even though these are imperfect measures of whether someone had a demanding job, they cover several aspects of demanding work. By evaluating the relationship between exemption use and responses to these questions, we can gain insights into which aspects of demanding work are associated with exemption use.

To evaluate the regulation, we will compare the estimated demanding work measures for exemption users to 1) a random sample from the working population and 2) a random sample from the working population in the same age group. We hypothesise that examining the relationship between the features discussed in the questions and RVU usage will provide insights into who used the rule and whether the intended goals are met. Since the exemption came into effect in 2021, response predictions are made for the year 2020, as all exemption users should have been part of the employed population at the time. As we are interested in comparing exemption users with individuals who could have used the regulation, only individuals who were part of the employed population in 2020 are selected for both comparison samples.

A problem with this procedure is that the prediction model is unlikely to produce correctly generated binary labels for everyone. In other words, our proxies for demanding work are likely to be subject to some degree of misclassification. There are methods that account for this, but to yield correct results, knowledge of misclassification rates is required. Under the assumption that misclassification rates are nondifferential between exemption users and each comparison group, these rates can be estimated by holding out a subset of the respondents during training and evaluating prediction performance on them. However, transferability of these misclassification rates onto the exemption users requires assumptions that we argue are unlikely to hold.

1.3 Structure of this thesis

The remainder of this thesis is structured as follows: Section 2 covers the RVU levy exemption and problems that arise around classifying jobs as "demanding" more in-depth. Section 3 goes more in-depth into the statistical problems that arise. Section 4 describes the available data and performs some preliminary analyses. Section 5 discusses the prediction model and its performance. Section 6 uses the generated predictions to investigate what claims can be made about the performance of the regulation and how robust these claims are. Section 7 concludes.

2 Background

2.1 Demanding jobs

Bakker and Demerouti (2007) define demanding jobs as "jobs that require sustained physical and/or psychological effort". However, this definition of what makes a job "demanding" is still subject to some degree of subjectivity, and the Dutch legislature has struggled to consistently categorise jobs accordingly. This brings along the struggle of reaching the right audience with the right policy. For example, Zwick et al. (2022) show that an early retirement scheme for women in Germany did not reach the workers for whom it was intended, namely those with highly demanding jobs.

Hooftman and Houtman (2010) and van den Bogaard et al. (2016) argue that psychologically stressful and psychosocially demanding jobs should also be taken into account, not just physically demanding ones. The Federatie Nederlandse Vakbeweging (FNV) argues there are three types of demanding jobs:¹ 1) Physically demanding jobs that require heavy lifting and repetitive movements, 2) work that requires irregular work hours and night shifts, and 3) mentally demanding work that is emotionally heavy or with verbal abuse.

As early retirement is more common among lower-educated and working-class individuals, the abolition of the VUT and the discontinuation of other early retirement schemes can hit workers with demanding jobs harder (Visser et al., 2016). Vermeer et al. (2016) shows that the Dutch are willing to contribute towards pension plans for people with demanding work, enabling them to retire early, demonstrating that there is support among the population for early retirement schemes.

2.2 The RVU levy exemption

In 2021, a new regulation was introduced that allowed for the possibility of tax benefits on early retirement. This temporary RVU levy exemption was aimed at helping people with demanding jobs retire early. We will introduce the exemption step by step by answering three questions.

What is an RVU?

An early retirement scheme (regeling voor vervroegde uittreding, RVU) is an arrangement intended to bridge the financial gap between work and the SRA. That is, the employer pays the employee, either at once or periodically, from when they stop working until the SRA. The arrangement can be made either individually between an employer and an employee, or collectively through a collectieve arbeidsovereenkomst (cao). The conditions for qualification as RVU, judged on objective conditions and features, not the motives of either the employer or employee, are twofold: 1) The dismissal has to be age-related, not dysfunction, reorganisation, etc., and 2) the payment has to be large enough such that it can sustain the employee until retirement,

¹<https://www.fnv.nl/werk-inkomen/duurzame-inzetbaarheid-eerder-stoppen-met-werken>

determined by the 70%-toets. Details of the requirements can be found in Artikel 32ba Wet LB.²

What is the RVU levy?

In a regular situation, when an RVU is agreed upon, a pseudo final levy has to be paid over the agreed amount. The applicable levy rate was 52% in 2025 but will gradually increase to 65% in 2028.³ This levy, known as the RVU levy, is meant to discourage people from agreeing on an RVU and keep people in the workforce longer (bron).

What is the RVU levy exemption?

The exemption threshold Early Retirement Scheme (drempelvrijstelling RVU-heffing), or RVU levy exemption, is a temporary relaxation of the RVU levy introduced on January 1st, 2021. This threshold exemption means the levy does not have to be paid over amounts up to the equivalent of the net AOW amount (€2273 in 2025 and €2,657 in 2026) for at most 36 months. Originally, the exemption was set to last until 2025, but it has been extended until the end of 2028 under the condition that it will be evaluated periodically.

3 Theoretical framework

3.1 Entities of interest

Let Y be a binary variable denoting whether someone made use of the RVU levy exemption and X be whether someone had a demanding job.

	$Y = 0$	$Y = 1$
$X = 0$	n_{00}	n_{01}
$X = 1$	n_{10}	n_{11}

Table 1: $P(X, Y)$

	$Y = 0$	$Y = 1$
$X = 0$	$1 - \alpha_0$	$1 - \alpha_1$
$X = 1$	α_0	α_1

Table 2: $P(X|Y)$

	$Y = 0$	$Y = 1$
$X = 0$	$1 - \pi_0$	π_0
$X = 1$	$1 - \pi_1$	π_1

Table 3: $P(Y|X)$

We are mainly interested in the probability that an RVU user had demanding work,

$$P(X = 0|Y = 1) \text{ vs } P(X = 1|Y = 1),$$

and whether individuals with demanding work are over- or underrepresented among RVU users

$$P(X = 1|Y = 1) \text{ vs } P(X = 1|Y = 0).$$

Following the notation of Buonaccorsi (2010), the former is evaluated by looking at

$$\alpha_y = P(X = 1|Y = y) = 1 - P(X = 0|Y = y),$$

²A summary can be found [here](#).

³<https://business.gov.nl/subsidies-and-schemes/early-retirement-scheme-rvu/>

which can be estimated by

$$a_y = \frac{n_{1y}}{n_{0y} + n_{1y}}, \quad (1)$$

the fraction of users that had a demanding job given Y , and the latter is evaluated by looking at the odds ratio

$$\Psi = \frac{a_1(1 - a_0)}{a_0(1 - a_1)}, \quad (2)$$

where $\Psi > 1$ indicates $a_1 > a_0$. The odds ratio is symmetric in the sense that switching the values of a_1 and a_0 is the same as inverting the odds ratio. Also, the odds ratio is easy for hypothesis tests.

In our setting, we pre-stratify on exemption usage, sampling independently from the control group consisting of individuals that did not use the exemption. This case-control setting simplifies comparisons with similar groups, but it prevents us from investigating the fraction of individuals with demanding work who use the exemption. Formally, this setting allows us to make statements about α , as specified in Table 2, and functions thereof, but it prevents us from doing the same about π , as in 3.

One might suggest that a regression of the form $Y = \alpha + \beta X + \gamma Z$ would be appropriate to account for the effect of potentially relevant covariates Z , but we argue that the interest lies more in population shares rather than the "causal" effects of demanding work.⁴

Because exemption use is restricted only to employees in the last three years before the SRA, one might not only be interested in using the full employed population as control sample, but also the population eligible for exemption use. As not all individuals make use of the full 3 years, some ages appear more frequently among exemption users. To account for this, we construct a comparison sample where each age is represented as frequent as in exemption usage.⁵

More formally, let C denote inclusion in a subsample such that $P(\text{age}|Y = 0, C = 1) = P(\text{age}|Y = 1)$, which indicates whether an individual is in the same age group as exemption users. Then we are also interested in

$$P(X = 1|Y = 1, C = 1) \text{ vs } P(X = 1|Y = 0, C = 1),$$

where $P(X = 1|Y = 1, C = 1) = P(X = 1|Y = 1)$ holds trivially as all individuals that made use of the exemption ($Y = 1$) are in that age group. $P(X = 1|Y = 0, C)$, on the contrary, might be differential respective to C as individuals probabilities of having demanding jobs might differ across ages.

⁴As an example, imagine a simplified and unrealistic situation where 90% of the individuals in the construction sector can have demanding jobs and no individuals outside of the sectors do. Additionally, all individuals in the construction sector make use of the RVU and none outside do. Then, running a regression with sector as covariate would attribute all explanatory power to the sector and none to the demanding work variable, implying the regulation does not work effectively, whereas it reaches all individuals with demanding work and 90% of the individuals who use the RVU have demanding work.

⁵In our study we stratify on both gender and age. Severe computation times of multiples days prevented us from rerunning the experiment with a different control sample. We consider this a limitation of our study.

3.2 The misclassification problem

The apparent problem that arises is that there is no directly available data on demanding jobs for exemption users. That is, we know the number of people that made use of the exemption, $n_{11} + n_{01}$, but we do not know n_{11} and n_{01} themselves. Hence we can not estimate α_1 directly.

To estimate these conditional probabilities, we generate predictions for several aspects associated with demanding work, denoted by $\hat{X}$, based on microdata features including age, wage, education level, and gender, denoted by Z . That is we train a prediction model such that $\hat{X} = f(Z)$, where $f(\cdot)$ is some prediction function.

$\hat{X}$	Y		$\hat{X}$	X		$\hat{X}$	X	
	0	1		0	1		0	1
0	$\dot{n}_{0,0}$	$\dot{n}_{0,1}$	0	$\ddot{n}_{0,0}$	$\ddot{n}_{0,1}$	0	$\theta_{0 0,Y}$	$\theta_{0 1,Y}$
1	$\dot{n}_{1,0}$	$\dot{n}_{1,1}$	1	$\ddot{n}_{1,0}$	$\ddot{n}_{1,1}$	1	$\theta_{1 0,Y}$	$\theta_{1 1,Y}$
(a) $\dot{n}_{\hat{X},Y}$			(b) $\ddot{n}_{\hat{X},X}$			(c) $\theta_{\hat{X} X,Y}$		

Table 4: Counts and conditional probabilities relating predictions $\hat{X}$, true states X , and outcomes Y .

Imperfection of these predictions implies that our demanding work proxies are subject to misclassification. Let $\theta_{\hat{X}|X} = P(\hat{X} = \hat{x}|X = x)$ denote the probability of the demanding work prediction being equal to $\hat{x} \in \{0, 1\}$, given that the actual response is (or would have been) $x \in \{0, 1\}$. Then, the probabilities of correct classification $\theta_{1|1}$ and $\theta_{0|0}$, respectively referred to as the sensitivity (Se) and specificity (Sp), are both strictly smaller than one. Equivalently, the misclassification probabilities $\theta_{1|0}$ and $\theta_{0|1}$ are nonzero.⁶

A naive estimate of α that does not take misclassification into account is given by

$$a = \frac{\dot{n}_{i,i}}{\dot{n}_{i,i} + \dot{n}_{(1-i),i}} \quad (3)$$

The sensitivity and specificity can be estimated by

$$\hat{\theta}_{i|i} = \frac{\ddot{n}_{i,i}}{\ddot{n}_{i,i} + \ddot{n}_{(1-i),i}} \quad (4)$$

In addition, it is non-trivial that misclassification probabilities are equivalent between sub-populations. That is, $\theta_{1|1}$ and $\theta_{0|0}$ may differ between certain groups. For example, let $\theta_{\hat{X}|X,Y} = P(\hat{X} = \hat{x}|X = x, Y = y)$ be the probability of classifying demanding work as $\hat{x} \in \{0, 1\}$ given the true value $x \in \{0, 1\}$ and exemption use $y \in \{0, 1\}$, then it could be that

⁶As we assume the true X is fixed, we are looking at misclassification. In a setting where the observed is fixed, following Buonaccorsi (2010) notation, the entities of interest are reclassification probabilities $\lambda_{X|\hat{X},Y} = P(X = x|\hat{X} = x, Y = y)$.

$\theta_{\hat{X}|X,1} \neq \theta_{\hat{X}|X,0}$ and misclassification rates are differential with respect to Y . Here, they are differential in the sense that they differ between exemption users and non-exemption users.

Opposed to differentiability, nondifferentiability assumes that misclassification rates are equivalent between groups. In the situation of nondifferentiability of misclassification probabilities with respect to exemption usage, we can write

$$P(X|\hat{X}, Y) = P(X|\hat{X}),$$

or equivalent $\theta_{\hat{x}|x,y} = \theta_{\hat{x}|x}$.

3.3 Correcting for misclassification

Presence of misclassification means $\hat{X} \neq X$ and hence $P(Y|X) = P(Y|\hat{X})$ does not hold trivially. Therefore, naively estimating α_y by

$$a_y = \frac{\dot{n}_{1,y}}{\dot{n}_{1,y} + \dot{n}_{0,y}},$$

where $\dot{n}_{\hat{x},y}$ is the number of individuals with demanding work prediction $\hat{x}$ and exemption usage y , yields a biased estimate (Buonaccorsi, 2010).

There are several methods that appropriately account for measurement error, where the suitability of each method depends on the available data and the problem at hand (see Buonaccorsi (2010) for an introduction and overview of several methods). In the setting of this thesis, we are mainly interested in the relation between two categorical variables, exemption usage Y and aspects of demanding work X , which is referred to as misclassification in two-way tables.

In this setting, correcting for misclassification is straightforward if misclassification probabilities are known. Following Buonaccorsi (2010), we specify a corrected estimate of α_y as

$$\hat{\alpha}_y = \frac{a_y - P(\hat{X} = 1|X = 0, Y = y)}{P(\hat{X} = 1|X = 1, Y = y) - P(\hat{X} = 1|X = 0, Y = y)} = \frac{a_y - (1 - \theta_{0|0,y})}{\theta_{1|1,y} + \theta_{0|0,y} - 1}, \quad (5)$$

where a_y is the naive estimate of α_y , and $\theta_{1|1,y}$ and $\theta_{0|0,y}$ again denote the sensitivity and specificity, respectively, given $Y = y$. Section 3.5 briefly discusses the intuition behind this formula.

For the population of survey respondents, the sensitivity and specificity can be adequately estimated by holding out a random test set on which the prediction model is not trained but only tested. However, as the true responses of exemption users and individuals in the control populations are generally unavailable, we can not estimate these directly. Additionally, as misclassification probabilities need not be transportable between the survey respondents and other populations, nondifferentiability need not hold by definition and assumptions have to made

in order to perform inference.

According to the LISS panel, their respondents are a representative sample of the Dutch population. Hence, the subsample that are part of the employed population should in theory be a representative sample of the employed population. Moreover, given enough observations, each subsample of employed individuals with a certain age should be a representative sample of the employed population with that age. If this claim were true, the survey respondents and control samples should (given age) be drawn independently from the same population, and we argue that the assumption of nondifferentiability would hold. As a result, the misclassification probabilities estimated on the test set would be transportable to the comparison sets and valid inference could be performed. Section 4 investigates this claim. However, by assuming that exemption users are a random sample of the employed population, we implicitly assume they are a random sample of individuals with demanding jobs. The next section discusses implications of this in more detail.

3.4 Prediction for demanding work

In our setting, we predict whether individuals have demanding work based on prediction model $f(\cdot)$ and features Z . Moreover, the predicted values $\hat{X} = \hat{f}(Z)$ are a deterministic function of the features. Training the model comes down to learning about the relation $P(X|Z)$ to find which regions of Z should map to $\hat{X} = 0$ and which to $\hat{X} = 1$. That is, we train $f(\cdot)$ to find an "optimal" A_i such that $f(z) = i \forall z \in A_i, i \in \{0, 1\}$.

Since we are interested in the sensitivity and specificity of the predictor, our definition of "optimal" is not simply the prediction accuracy $P(\hat{x} = x)$, but we rather look at the underlying distribution $P(\hat{X}|X)$. To account for class imbalance of X , we aim to find $f(\cdot)$ that yields the highest Youden's J, which is equivalent to finding the regions $A_i = \{z \in \mathcal{Z} : P(z|X = i) > P(z|X = 1 - i)\}, i \in 0, 1$, which maximize the mean of the sensitivity and specificity. This is attained by setting $f(z)$ equal to the X that has the highest density for that z . A simplified visualisation of this can be found in Figure 1.

Given the trained predictor, predictions for new samples are made by predicting $\hat{x} = 1$ if $z \in \hat{A}_1$ and $\hat{x} = 0$ otherwise. To avoid overestimating prediction performance as a result of overfitting, it is important to evaluate the performance on an independent test set. The simplified visualisation in Figure 2 shows how prediction probabilities θ can be estimated by calculating the fraction of each group that appears in $\hat{A}_i, i \in \{0, 1\}$.

In this process, misclassification of predictions can appear due to multiple reasons. First, unless the Z regions of $X = 0$ and $X = 1$ are completely separated, even the best possible deterministic prediction model of the form $\hat{X} = f(Z)$ produces misclassified predictions. Here, misclassification probabilities depend on the fraction of the conditional density of $Z|X = i$ that appears in the area where $P(Z|X = i) < P(Z|X = 1 - i)$. In other words $\hat{\theta}_{i|(1-i)0} = \int_{z \in \mathcal{Z}} \mathbb{1}_{\{z \in A_{1-i}\}} P(z|X = i) = \int_{z \in A_{1-i}} P(z|X = i), i \in \{0, 1\}$

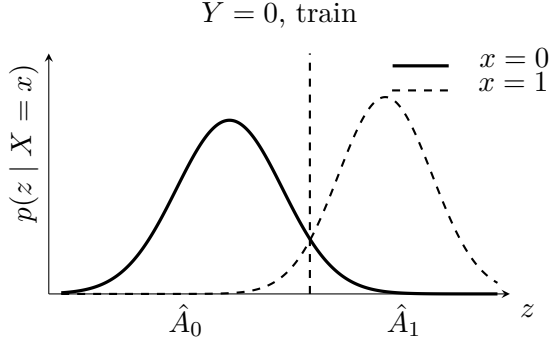


Figure 1: Training the predictor yields $\hat{A}_0$ and $\hat{A}_1$.

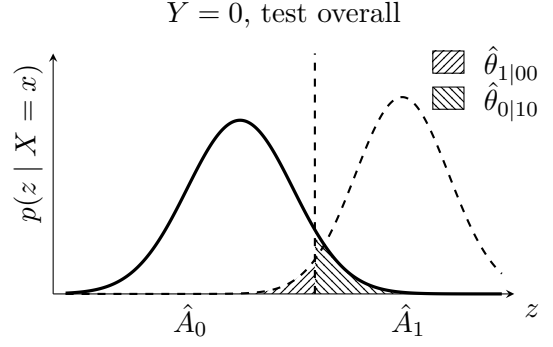


Figure 2: Evaluating performance on test set gives $\hat{\theta}$. Imperfectness of the decision boundary is due to approximation and estimation error.

Secondly, the above scenario where our trained model $\hat{f}(\cdot)$ finds the optimal $A_i, i \in \{0, 1\}$ is unlikely. As we do not have infinite data, 1) the limited training data will not contain enough information to learn the optimal regions, resulting in estimation error, and 2) as a result of sampling error, the train sample will not be a perfect representation of the full population. The latter can also be seen in the imperfect decision boundary in Figure 2.

Lastly, some models suffer from approximation error due to restrictions in functional form, which prevent the model to form the optimal regions even if infinite data were available. However, this is more so a problem for traditional econometric models rather than recent, more flexible, ML models.

3.5 Nondifferentiability of prediction errors

Looking at Figure 2, it is easily explained why the corrected estimate of a_y makes sense intuitively. Using $P(\hat{X} = 1|X = 1, Y = 0) = \theta_{1|10}$ and $P(\hat{X} = 0|X = 0, Y = 0) = \theta_{0|00}$, the probability that an individual will be classified as $\hat{X} = 1$ is equal to

$$P(\hat{X} = 1|Y = 0) = P(X = 1|Y = 0)\theta_{1|10} + P(X = 0|Y = 0)(1 - \theta_{0|00}), \quad (6)$$

which is a very intuitive formula. Rewriting this equation gives

$$P(X = 1) = \frac{P(\hat{X} = 1) - (1 - \theta_{0|00})}{\theta_{1|10} + \theta_{0|00} - 1}, \quad (7)$$

which corresponds to the corrected estimator in Eq. 5.

As this explanation unveils again, an accurate approximation of the misclassification rates of predictions is needed for valid inference. For subsamples with known true value, the model performance can be accurately estimated by comparing predictions against this true value. It is also possible to estimate the performance for a subsample of the test set, like for a specific age

group. See Figure 3 for a simplified example of how prediction errors can differ when stratifying the test sample on age.

However, for individuals in groups (2) - (4) who did not take the LISS survey, the true values are not known. As Figure 4 demonstrates, directly estimating misclassification rates of the predictions requires either more data or assumptions.

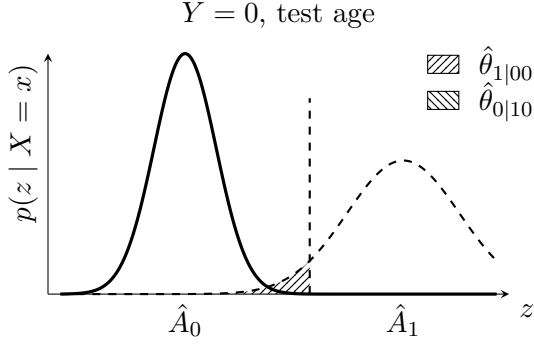


Figure 3: Evaluating performance for age-stratified sample gives alternative $\hat{\theta}$.

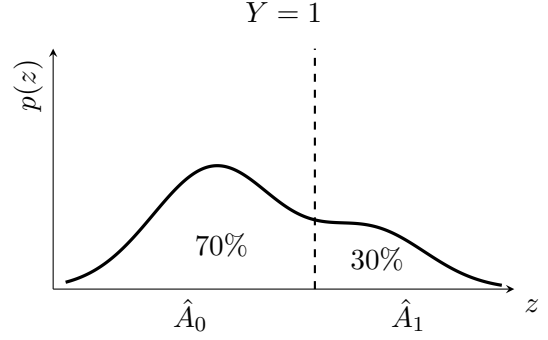


Figure 4: For groups (2) - (4), true values of x are not known.

As mentioned above, the probability of predictions being misclassified can be written as $\theta_{i|(1-i)0} = \int_{z \in A_{1-i}} P(z|X=i)$, $i \in \{0,1\}$. Using this formulation, the nondifferentiality assumes requires

$$\int_{z \in A_{1-i}} P(z|X=i, Y=0) = \int_{z \in A_{1-i}} P(z|X=i, Y=1), \quad i \in \{0,1\}, \quad (8)$$

which says the density of z given x must have the same weight in regions A_i across Y . This can be true in two ways: 1) $P(Z|X,Y) = P(Z|X)$, they have the same distribution, which is a sufficient condition for Eq. 8 to hold, or 2) $P(Z|X,Y) \neq P(Z|X)$, but the weight is distributed such that Eq. 8 still holds..

We argue that the former could potentially hold structurally under restrictive assumptions, but the latter is a very unlikely assumption to hold structurally, as we hypothesize it could only reasonably hold due to chance.

Namely, the former implies that given demanding work, exemption usage is a random sample of individuals. In this case, differences in $\hat{X}$ between $Y=0$ and $Y=1$ are only due to differences in $P(X)$. However, also this assumption is unlikely to hold, as we hypothesize it is unlikely that each individual with a demanding job had the same probability of using the exemption (for example in specific sectors).

Concluding, we hypothesize that the nondifferentiality assumption is unlikely to hold 1) because the exemption users could be sampled from a region of Z where performance is different from the comparison sample and 2) because even given Z , RVU users could be a nonrandom sample with different underlying relation (for example due to unobserved variables). Both could lead to

different classification performance

We can however, be confident in ruling out some combinations of sensitivity and specificity. For example in Figure 4, following Eq. 6, a specificity of 50% and a sensitivity of 100% would be highly unlikely to yield 30% of observations as $\hat{x} = 1$, as this combination would be expected to produce at least as many prediction for $\hat{X} = 1$ for any $P(X) \in [0, 1]$.

4 Data

This thesis uses three data sources: 1) a list of individuals who made use of the RVU levy exemption, 2) responses to the LISS Work & Schooling survey, and 3) registration data on individuals which can be accessed in the CBS microdata. All datasets were uploaded to/accessed in the CBS Microdata Remote Access (RA) environment.

4.1 CBS Microdata

The CBS microdata contains many detailed individual-level anonymized datasets about the Netherlands. This ranges from personal records like birth months and salaries to data about households and organizations. Use of this data is under strict confidentiality and only available for approved research institutions. To guarantee privacy, raw datasets can only be accessed in the secure RA environment and results have to satisfy strict anti-disclosure criteria before they can be exported from the secure environment. One of these criteria is that cells in tables are not allowed to be based on fewer than 10 individuals, so to comply with these standards, some tables our with results contain merged cells.

In this research, we make use of the following GBA, SPOLISBUS, and HOOGSTEOPLTAB microdatasets. The GBA (gemeentelijke basisadministratie) contains information on gender and birth year, the SPOLISBUS contains monthly data from the "polisadministratie" which covers salary, hours worked, whether the individual had a car from the employer, sector, and extra hours worked. Lastly the HOOGSTEOPLTAB contains data on the highest achieved education.

4.2 RVU users

To investigate who used RVU levy assumption, we make use of a list of individuals who made use of an RVU. These individuals appear with code 53 (early retirement scheme) as type of employment relationship in the SPOLISBUS. Since this data is not directly available in the regular SPOLISBUS, it was provided to us by the UWV.

This datasets contains 36,414 individuals in total. Each individual is labelled either "RVU only in 2021 or later" or "Likely already before 2021". In total, 27,016 get the former label and 9,398 the latter. As the RVU levy exemption only entered force in 2021, the group with the latter label

already went with an early retirement scheme before the exemption came into effect. Hence, only the former group is taken into consideration as exemption users in this study.

4.3 LISS Work & Schooling survey

As proxies for aspects associated with demanding work, we make use of several questions of the LISS Work & Schooling survey. This yearly survey consists of 17 waves (2008 - 2024) and the number participations per respondent ranges from 1 tot 17. According to the LISS panel, their surveys are representative samples of the Dutch population (bron).

Throughout the waves, questions have generally stayed the same or similar. However, sometimes questions were changed. Each survey had around many questions covering a wide range of topics related to work and schooling. Not all respondents had to answer all questions, as previous answers impact which questions one has to answer. From these questions, we identified six questions as most relevant regarding demanding work. These are,

- (a) **Profession:** What is you current profession / What profession did you exercise in your last job?
- (b) **Irregular hours:** Do/did you work irregular hours?
- (c) **Physically demanding (agree):** My job is/was physically demanding.
- (d) **Physically demanding (often):** Is/was your job physically demanding?
- (e) **Heavy lifting:** Do/did you need to lift heavy objects?
- (f) **Kneel / stoop:** Do/did you need to kneel or stoop?

where questions (b), (d), (e), and (f) were answered with never/sometimes/often, question (c) with strongly disagree/disagree/agree/strongly agree, and question (a) with nine answers which we divided into the categories mental/manual/supervisory. These responses are turned into a binary indicator X , where $X = 1$ is associated with work being demanding. Answers were classified as demanding if the response is often, agree/strongly agree, or supervisory.

Here, question (a), whether someone had a supervisory profession, might appear as a weird question. We include it as the preliminary analysis of the overlap between exemption users and LISS respondents showed they were strongly overrepresented in the overlap, compared to LISS respondents who did not take an RVU.

It is important to note that this survey consists of self-reported data and is hence likely to contains measurement errors as well. However, to not further increase the complexity of the analysis, we do not take this into account in the analysis. Out of 99,404 responses across different waves, 13,896 belonged to individuals who did not give permission to connect to the microdata, 1,768 to

individuals who could not be found in the microdata, and 83,740 responses were connected.

4.4 Comparison groups

To evaluate the RVU levy exemption, we make use of four comparison groups within the employed population:

- (1) LISS survey respondents
- (2) Exemption users
- (3) Non-exemption users
- (4) Age-matched non-users

Groups (1) and (2) are predefined, as we can not influence participation in the LISS survey and exemption usage. The small sample of individuals who appear in the LISS survey and are also exemption users are assigned to group (1), as they are assumed to be a random sample of (2), but not a random sample of (1). Hence, assigning them to (2) would induce a selection bias in (1).

Since the exemption can only be used by individuals in the employed population, groups (3) and (4) are constructed by randomly selecting individuals from the GBA who appeared in the SPOLIS 2020. For group (4), a weighted draw is done where weights are relative to the number of exemption users with the same birth year. It should be noted that these two groups are a random sample of the employed non-user population rather than just the employed population. This subtle difference results in that exemption users are underrepresented in comparison groups (3) and (4)

Table 5 shows the original sizes of each group and the number of individuals that appear in the SPOLIS and 2020 edition of the SPOLIS, respectively. Remarkable is the appearance number of comparison group (2), as apparently not all individuals who made use of an RVU appeared in the SPOLIS between 2010 and 2025, and a significantly larger part did not appear in the 2020 SPOLIS.

Group	n	n_{SPOLIS}	$n_{\text{SPOLIS},2020}$
(1)-(4)	200,055	129,019	93,362
(1)	13,059	10,441	7,745
(2)	27,016	27,006	26,618
(3)	100,000	51,969	31,483
(4)	60,000	39,623	27,535

Table 5: Number of individuals that appear in SPOLIS, by comparison group.

As argued in Section 3, inference validity depends on whether misclassification probabilities are nondifferential between comparison groups. In turn, nondifferentiability is influenced by whether the comparison groups are randomly sampled (given age) from the overall employed population. Figures 5 and 6 show the distributions of several characteristics for the comparison groups.

For comparison, survey respondents are filtered for appearance in the 2020 SPOLIS, but to increase the sample size while training the ML model, individuals are kept if they appear at least once in the SPOLIS between 2010 and 2025.

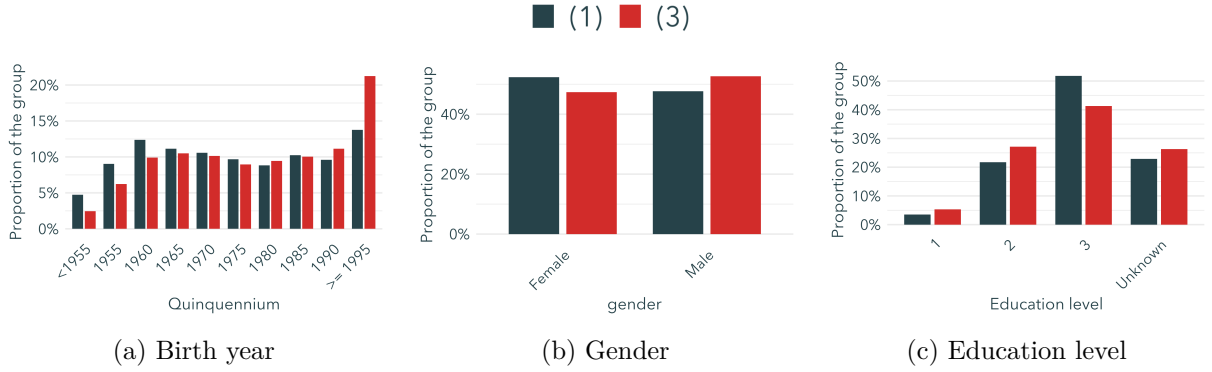


Figure 5: Comparison between groups (1) and (3).

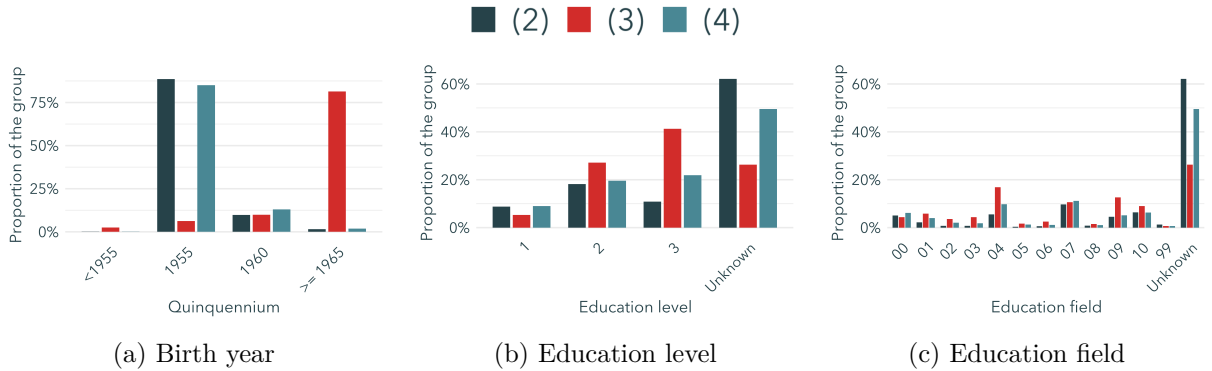


Figure 6: Characteristics distributions of comparison groups (2), (3) and (4).

Figure 5 shows that the distribution of certain characteristics do differ between survey users are the random sample of the employed non-users. The youngest age cohorts appear more often in the random sample and the oldest appear more often in the respondents sample. Also, females are slightly overrepresented in the LISS survey, as well as individuals with a high education level.

Figure 6 shows the distributions of several characteristics of comparison groups (2) - (4). It is apparent that there are large differences between the random group (3) and groups (2) and (4). Additionally, there are also significant differences between groups (2) and (4), suggesting that the exemption users have a different Z -distribution and hence it is unlikely for the assumption of nondifferentiability to hold.

5 Predictions

We use Machine Learning to predict what responses to demanding work-related survey questions would have been, had the comparison groups been asked the questions. See 4 for a list of these questions used. We train and evaluate two models, a random forest and XGBoost, on group (1) for which the true answer is known. Using the trained models we make predictions for the individuals in groups (2) - (4).

In this procedure, it is important to note that sheer prediction performance is not the most important entity. As discussed in Sections 1 and 3, an accurate estimate of the prediction performance, specifically the sensitivity and specificity, is more important as the proposed methodology is able to account for misclassification if misclassification rates are known.

5.1 Machine Learning models

The machine learning models used are random forest and XGBoost. The main advantage of these models compared to more traditional econometric models is that they are more flexible as they do not impose a linear relation between the features and outcome variable. We hypothesize that this flexibility leads to a more accurate estimation of $P(X|Z)$ across regions of Z (Section intuition). Also, we are just interested in prediction performance, not in the relation between features Z and outcome variable X . Therefore it is not a problem that the ML models are black boxes. Both models are implemented in R.

5.2 Features

We construct features from the CBS microdata. As both models are flexible and can generally select important variables themselves (bron), we use many features. In addition to features already available in the microdata, we construct some features ourselves.

For the SPOLIS features, monthly entries are brought back to the first month the individual appears in. To account for potential within-year differences, new features are constructed such as the number of simultaneous contracts, the variation of salary within the year, and the variation in hours worked within the year. Additionally, information about the individuals is used such as gender, highest achieved education level and corresponding education field, and age.

As data from many years is available, both in the microdata and LISS survey, a target year is chosen for which the features are computed and outcome is predicted. SPOLIS features are taken at several time windows with respect to the target year (e.g. 10 years before, one year before, year of, 1 year after), and age is computed for the target year.

For group (1) the target year is equal to the year the survey was filled out if the person was working at the time, or the last year the person worked if they were not working anymore. For

comparison groups (2) - (4), the target year is set to 2020.

5.3 Training & tuning

As the target variables are only available for the LISS respondents, the model is trained and evaluated only on this sample. To this end, the sample is split in three groups: 52.5% training, 17.5% validation, 30% test. The test set is relatively big, as the emphasis is on getting accurate estimates of performance. Here the split is based on individuals rather than responses in order to ensure there is no spillover during training/testing. The test set is only used once to avoid overfitting and produce accurate performance estimates.

Both models produce a score between 0 and 1 corresponding to the estimated likelihood of the response being equal to 1. As our methodology requires the estimates to be binary, these scores are converted to 0/1 based on a threshold. Increasing the value of this threshold will lead to fewer individuals being classified as 1 and hence inherently increasing the specificity and decreasing the sensitivity. Reducing the threshold has the opposite effect.

The threshold is chosen which maximizes Youden's J statistic,

$$J = \text{sensitivity} + \text{specificity} - 1,$$

which is equivalent to maximizing the average of the sensitivity and specificity, as discussed in section 3

Additionally, during training the observations are given weights equal to the inverse relative frequency, such that each group gets the same total weight. Parameters are optimized through grid search based on J on a separate evaluation set. The best model, by J , per question is chosen to generate the labels for comparison groups (2) - (4).

5.4 Estimating the sensitivity and specificity

Table 6 contains the confusion matrices of the best prediction model for each question, evaluated on the test set. From these, following Eq. 4, the specificity can be estimated by

$$\hat{\theta}_{0|00} = \frac{\ddot{n}_{0,0}}{\ddot{n}_{0,0} + \ddot{n}_{1,0}},$$

and the sensitivity can be estimated by

$$\hat{\theta}_{1|10} = \frac{\ddot{n}_{1,1}}{\ddot{n}_{1,1} + \ddot{n}_{0,1}},$$

where $\ddot{n}$ is specified as in Table 4. The resulting values can be seen in Table 7.

As the sensitivity and specificity differ substantially across different age groups (See Tables A.1 -

$\hat{X}$			$\hat{X}$			$\hat{X}$		
X	0	1	X	0	1	X	0	1
0	8669	3181	0	8579	3334	0	6939	3769
1	825	1802	1	867	1697	1	1037	2732
(a) Profession			(b) Irregular hours			(c) Physically demanding (agree)		

$\hat{X}$			$\hat{X}$			$\hat{X}$		
X	0	1	X	0	1	X	0	1
0	8585	4243	0	9390	3884	0	8082	3945
1	384	1265	1	363	840	1	616	1834
(d) Physically demanding (often)			(e) Heavy lifting			(f) Kneel / stoop		

Table 6: Confusion matrices for predictions of questions (a) - (f) on the test set. X denotes the true value and $\hat{X}$ denotes predictions.

A.6) we generate another estimate of the sensitivity and specificity, based on an age-stratified sample of the test set. More specifically, we compute weighted values, where each individual gets a weight equivalent to the fraction of exemption users with the same birth year.⁷ As mentioned in 4, we hypothesize that this age-stratified estimate could be a good approximation for comparison group (4) and a better approximation for group (2). The estimated specificities and sensitivities can be found in Table 7, where $\hat{\theta}_{i|i,all}$ refer to the estimates based on the full test set and $\hat{\theta}_{i|i,age}$ based on the age-stratified test set.

Q	$\hat{\theta}_{1 1,all}$	$\hat{\theta}_{0 0,all}$	$\hat{\theta}_{1 1,age}$	$\hat{\theta}_{0 0,age}$
(a)	0.686	0.732	0.782	0.685
(b)	0.662	0.720	0.670	0.755
(c)	0.725	0.648	0.706	0.633
(d)	0.767	0.669	0.824	0.710
(e)	0.698	0.707	0.712	0.755
(f)	0.749	0.672	0.750	0.685

Table 7: Estimates of the sensitivity and specificity, unweighted and age-weighted, for each question (a) - (f).

Figure 7 clearly shows that performance differs significantly between questions and between

⁷Ik besef nu ook dat dit nergens op slaat, als een bepaalde leeftijd nu vaker voorkomt dan de andere in LISS dan weegt die zwaarder...

6 Evaluation

6.1 Does the RVU levy exemption work?

Figure 7 shows the prediction rates of demanding work for the different comparison groups.

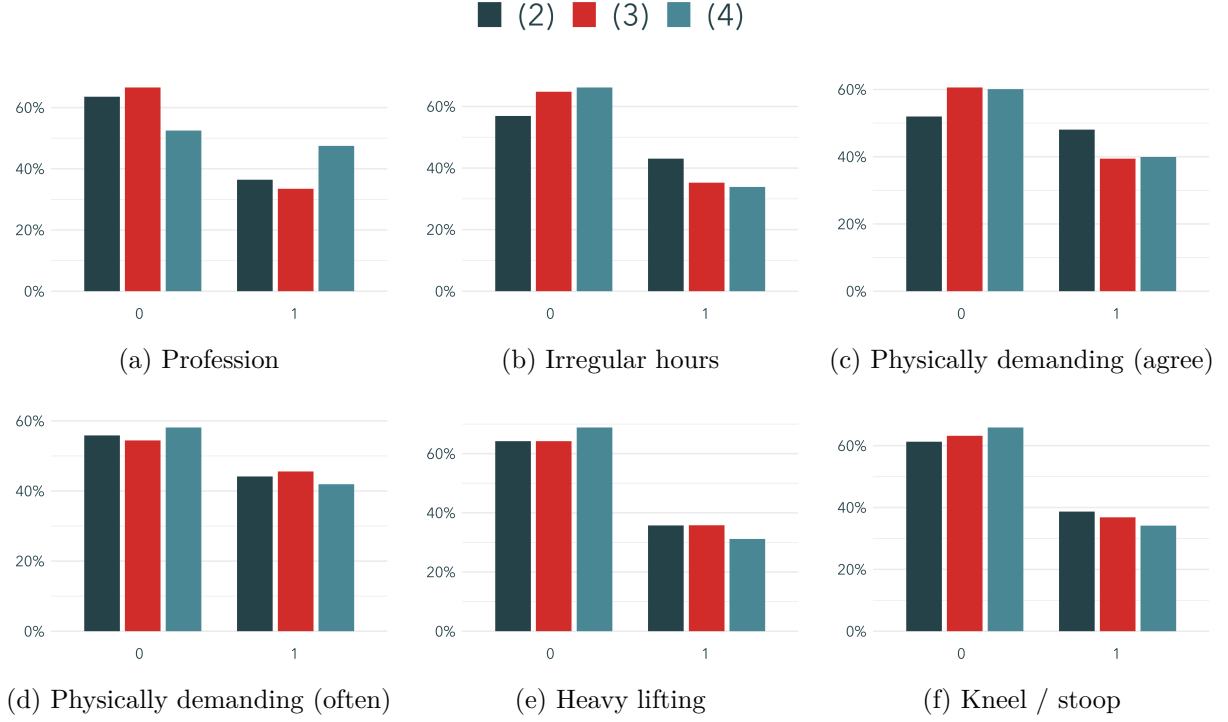


Figure 7: Prediction rates for each question (a) - (f) for comparison groups (2) - (4).

Following Eq. 3, a naive estimate of α_y is defined by $a_y = \frac{\hat{n}_{1,1}}{\hat{n}_{1,1} + \hat{n}_{0,1}}$, where $\hat{n}$ is defined as in Table 4, which is equivalent to the fraction of individuals with prediction $\hat{x} = 1$ within the group $Y = 1$.

However, to account for misclassification, we have to use the corrected estimate $\hat{\alpha}_y$, for which we need estimates of the sensitivity and specificity. As discussed in Section 5, we have two estimates of these, one for the overall test set and one for the age-matched test set.

For the sensitivity and specificity to be exportable to comparison groups (2) - (4), the nondifferentiability assumption of prediction errors with respect to Y is required. As discussed priorly, this assumption is unlikely to hold for group (2), but is more likely to hold for groups (3) and (4). For these two, we expect $\theta_{i|i,all}$ and $\theta_{i|i,age}$, respectively, could be close to the true values, as both groups (1), (3), and (4) are in some sense a random draw from the same population. Even though the preliminary analysis in Section 4 was not able to confirm this, continuing we assume these estimates are reliable.

For group (2), we have little confidence that the sensitivity and specificity are reliable estimates

of the true value. Section 6.2, therefore, investigates what would have been our conclusion for different estimates.

α shows the probability that the response of an exemption user to question Q would have been 1 or 0. Additionally, we are also interested in investigating whether these are over- or underrepresented within exemption users compared to both the full employed population and individuals who were eligible to use the exemption. Therefore, we also investigate the odds ratio Ψ , by estimating

$$\hat{\Psi}_c = \frac{\hat{\alpha}_1(1 - \hat{\alpha}_{0,c})}{\hat{\alpha}_{0,c}(1 - \hat{\alpha}_1)}, \quad (9)$$

where $c \in \{\text{age}, \text{random}\}$ are the corrected estimates of comparison groups (3) and (4) under the assumption of nondifferentiality with respect to the age-matched test set and full test set, respectively. Estimates of α and Ψ under the assumptions above can be found in Figure 8.

Q	$\hat{\alpha}_{1,\text{all}}$	$\hat{\alpha}_{1,\text{age}}$	$\hat{\Psi}_{\text{random}}$	$\hat{\Psi}_{\text{age}}$
(a)	0.23	0.11	0.63	0.23
(b)	0.40	0.44	3.32	2.75
(c)	0.34	0.34	3.94	4.77
(d)	0.25	0.28	0.99	1.24
(e)	0.16	0.24	1.65	1.91
(f)	0.14	0.17	1.87	3.08

Table 8: Estimates of α_1 and Ψ for each question (a) - (f).

Figure 8 shows our main results. $\hat{\alpha}_{1,c}$ shows our estimates of $P(X = 1|Y)$, the probability that an exemption user would have answered "often" to questions (b), (d), (e), and (f), "agree" or "strongly agree" to question (c), and "supervisory" to question (a), under the assumption of either nondifferentiality of prediction errors between the full test set ($c = \text{all}$) and the age-matched test subset ($c = \text{age}$).

These results indicate that in 2020, of exemption users, around

- (a) 10% - 25% had a supervisory profession,
- (b) 40% - 45% worked irregular hours,
- (c) 30% - 40% had physically demanding jobs,
- (d) 20% - 30% often had physically demanding jobs,
- (e) 15% - 25% needed to lift heavy objects,
- (f) 10% - 20% needed to kneel or stoop.

Additionally, individuals who worked irregular hours, had physically demanding jobs, needed to

lift heavy objects and had to kneel or stoop are overrepresented relative to both the full employed population and the exemption eligible population. Unlike the preliminary analysis suggested, individuals who had a supervisory profession are underrepresented, especially in comparison to their age cohort. Lastly, individuals whose job was often physically demanding appear slightly overrepresented in comparison to their age cohort, but neither under-nor overrepresented compared to the full unemployed population. This last results is interesting, particularly because of the similarity of questions (c) and (d).

Of course, once again, it should be stressed that these results strongly depend on the assumption of nondifferentiability, of which we argue the validity is unlikely.

6.2 Robustness

To investigate robustness of our results against the nondifferentiability assumption, Figure 8 shows what values the odds ratio would takes for different specifications of $\theta_{0|01}$, the specificity, and $\theta_{1|11}$, the sensitivity.

Figure 8 shows the odds ratios for different specification of θ . If the true, unknown, specificity and sensitivity are in the area above the red line, individuals with $X = 1$ are overrepresented among exemption users. If the true values are below the red line, they are underrepresented. The combination for which $\Psi < 0$ are very unlikely to contain the true values, as, argued in the 70/30 example of Section 3, these are unlikely to produce the observed prediction rates.

The blue and red dots denote the estimated specificity and sensitivity on the age-weighted and unweighted test sets, respectively. These give some indication of where the true values could lie, but they do not give any guarantee or measurable confidence. However, we hypothesize that the further the estimates are from the decision boundary of $\Psi = 1$, the more likely it is that $X = 1$ is overrepresented among exemption users.

It is apparent that, depending on the question, some areas associated with under- and overrepresentation cover close to the full likely set of values. For example, for question (b) to be underrepresented, the true sensitivity and specificity should lie in the small area. Using this argument, we argue that we can be confident in our representation conclusions for questions (c) and (f). For questions (a), (b) and (d), we are confident in our conclusion of representation in comparison to the exemption-eligible population.

7 Conclusion & discussion

This thesis aimed to asses whether the RVU levy exemption has been used by the intended target audience. To this end, we estimated the probability that the jobs of exemption users show several aspects related to demanding work. We did this predicting for each user what they would

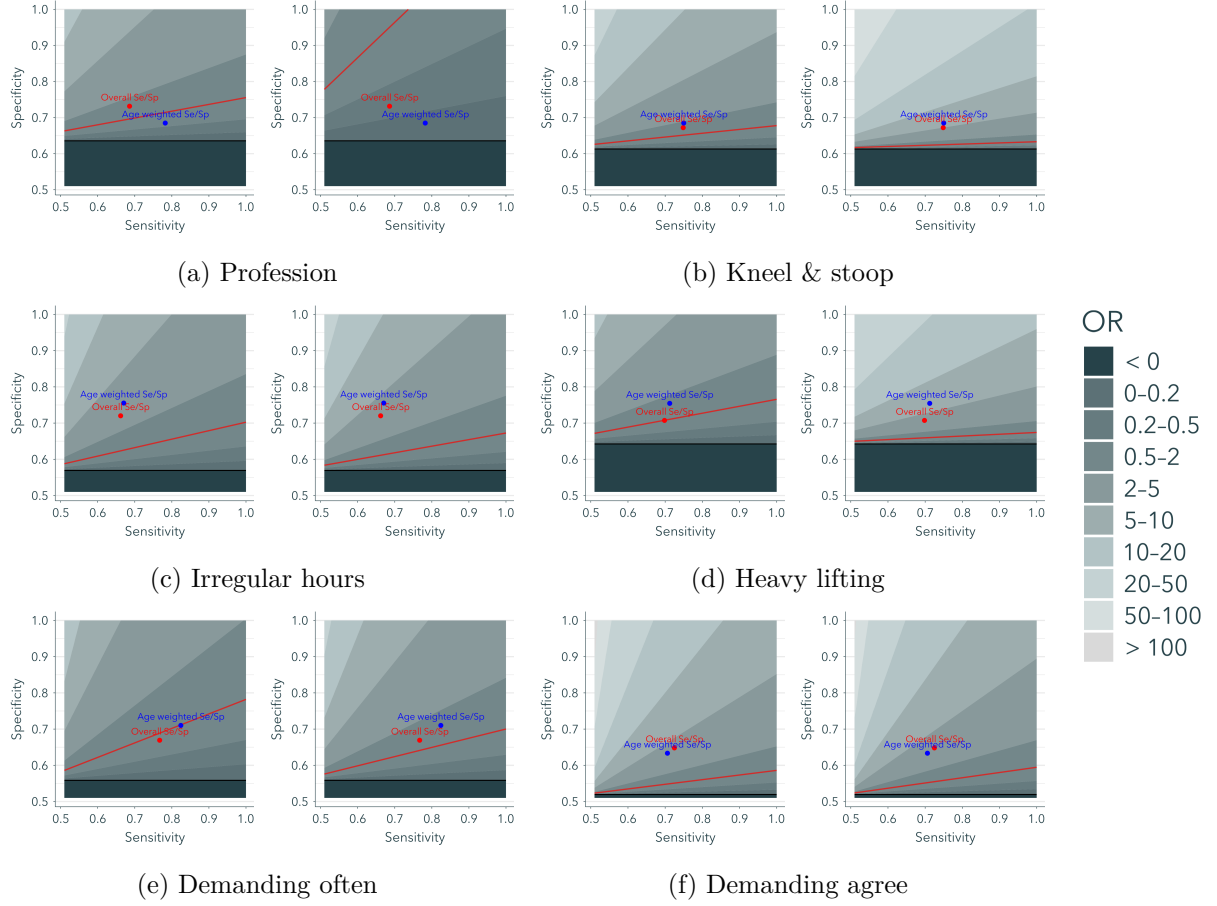


Figure 8: Odds ratios Ψ for different specifications of the specificity and sensitivity of the misclassified predictions of demanding work. For each pair of figures, the left evaluates the odds ratio against comparison group (3), and the right against comparison group (4). The red and black lines denote the decision boundary of $\Psi = 1$, and the unlikely boundary of $\Psi = 0$, respectively. Also, the blue and red dots denotes the age weighted and unweighted estimates of θ , respectively.

have answered to several questions in the LISS Work & Schooling survey, based on features in the CBS microdata. Using these predictions to perform inference, however, requires making the unlikely assumption of nondifferentiability of prediction error with respect to exemption usage.

If this nondifferentiability assumption is valid, a relatively large part of the exemption users had jobs associated with demanding work, and individuals who had demanding work are overrepresented among exemption users, both compared to the employed population as well as the exemption-eligible population. Investigating the robustness of these conclusions against the nondifferentiability assumption not holding shows that we can be confident in some conclusions, but care is required when making claims.

We conclude that our analysis suggests that the RVU levy exemption reached its intended target, in the sense that individuals who had demanding work appear to be **overrepresented among users**. However, our procedure of predicting survey responses inherently requires restrictive assumption, which are unlikely to hold in practice. To make confident claims about the performance of the RVU levy exemption, further analysis, and preferably more data, is required.

7.1 Discussion

Next to general downsides associated with using predicted labels mentioned throughout this thesis, this study has some severe limitations.

First, the data is not the most reliable. Our measures for demanding work, the LISS survey responses, are self-reported and hence likely to contain mismeasured values. This is not taken into account explicitly in our research. Additionally, the dataset of exemption users appears to be subject to some noise. **As it was derived from a code related to early retirement in the SPOLIS, it likely not a perfect measure of RVU levy exemption usage.** This argument is supported by some individuals not appearing in the SPOLIS who did take an RVU.

Second, next to the assumption of nondifferentiability of prediction errors between exemption users and comparison groups, we also make the assumption that misclassification rates are transportable from the LISS respondent onto the samples of the full and age-matched employed population, even though a preliminary analysis shows that this need **need** not be a valid assumption. This assumption is not tested and could impact the reliability of our results. Also, we only look at point estimates of the fraction of exemption users with demanding work, and do not take into account uncertainty associated with it.

Third, the discussion of what work classifies as demanding is still ongoing. We proxy demanding work by using responses to several questions related to demanding work, but these are neither perfect nor exhaustive.

Fourth, as access to the CBS microdata was revoked before the writing procedure started, we ran into several error during the writing process which could not be restored. While constructing

the comparison samples, I made several mistakes, such as stratifying on gender, introducing selection bias by selecting subsamples in the wrong order, and unnecessarily filtering for overlap in comparison sets. Also when computing the weighted sensitivity and specificity estimates, the weighting procedure does not work as intended.

7.2 Further research

It is evident that there is much room for further research into the RVU levy exemption. This thesis lays a strong foundation of methods that can be used when data is missing, and the problems encountered when using predictions, but there are many opportunities for continuation and improvements, starting with correcting fixing the errors described above since these are relatively straightforward to solve when you know what to look out for.

A main problem in this research is the absence of data. Where direct information on whether exemption users had demanding jobs would solve the problem at hand directly, we expect this is unlikely to hold. However, more information on a (small) subsample could already give more insights, for example by investigating methods that are able to work with internal validation samples.

There is much room for improvement in the prediction model. A more thorough preliminary analysis of what features correspond to exemption users could already improve prediction performance. Also, use of more (informative) features could yield improvements. We expect that, for example health data could significantly decrease misclassification rates. Also, more/other questions regarding to demanding work could be used, for example from the EBB, or by combining questions from the LISS survey.

In this study, close attention is paid to the fraction of individuals within exemption users who had demanding work and whether they are over or underrepresented, but the setup of our procedure prevented us from investigating how large a part of the population with demanding jobs made use of the RVU levy exemption. Other entities? It could also be of interest to investigate what fraction of the exemption users had demanding jobs more than on year before the exemption came into effect.

The prediction models produced values between zero and one, which were converted to binary indicators based on a threshold. We hypothesize that in this process some information was lost and using the raw predictions could be a direction of continuation, perhaps to improve estimates of the sensitivity and specificity. Lastly, we hypothesize that under some assumptions of prior distributions, the robustness analysis would be an interesting area to investigate from a Bayesian perspective.



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A Supplementary results

Table A.1: Sensitivity and specificity by birth cohort: `profession`.

Cohort	Sensitivity	Specificity
<1955	0.732	0.673
1955	0.814	0.680
1960	0.785	0.732
1965	0.771	0.651
1970	0.787	0.639
1975	0.665	0.678
1980	0.545	0.683
1985	0.439	0.770
1990	0.565	0.870
≥ 1995	0.165	0.936

Table A.2: Sensitivity and specificity by birth cohort: `job_extra_hours`.

Cohort	Sensitivity	Specificity
<1955	0.616	0.637
1955	0.690	0.737
1960	0.706	0.699
1965	0.778	0.675
1970	0.603	0.751
1975	0.685	0.740
1980	0.597	0.664
1985	0.761	0.667
1990	0.576	0.769
≥ 1995	0.433	0.863

Table A.3: Sensitivity and specificity by birth cohort: `job_phys_demand_often`.

Cohort	Sensitivity	Specificity
<1955	0.652	0.731
1955	0.853	0.699
1960	0.897	0.631
1965	0.719	0.618
1970	0.728	0.621
1975	0.766	0.666
1980	0.808	0.610
1985	0.803	0.587
1990	0.750	0.714
≥ 1995	0.479	0.821

Table A.4: Sensitivity and specificity by birth cohort: `job_phys_demand_agree`.

Cohort	Sensitivity	Specificity
<1955	0.645	0.626
1955	0.718	0.604
1960	0.796	0.590
1965	0.719	0.567
1970	0.794	0.602
1975	0.779	0.665
1980	0.740	0.610
1985	0.783	0.632
1990	0.612	0.749
≥ 1995	0.444	0.848

Table A.5: Sensitivity and specificity by birth cohort: `job_lift_heavy`.

Cohort	Sensitivity	Specificity
<1955	0.500	0.810
1955	0.712	0.745
1960	0.834	0.652
1965	0.713	0.642
1970	0.664	0.643
1975	0.744	0.709
1980	0.716	0.700
1985	0.772	0.672
1990	0.626	0.746
≥ 1995	0.352	0.779

Table A.6: Sensitivity and specificity by birth cohort: `job_kneel_stoop`.

Cohort	Sensitivity	Specificity
<1955	0.585	0.717
1955	0.759	0.675
1960	0.865	0.638
1965	0.746	0.611
1970	0.797	0.642
1975	0.828	0.678
1980	0.811	0.632
1985	0.757	0.613
1990	0.628	0.736
≥ 1995	0.443	0.784